Separating Basis and Operator Conditioning in Plane-Wave DG and the Ultra-Weak Variational Formulation

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Abstract

Plane-wave Trefftz discretizations can be extremely ill-conditioned in coefficient coordinates even when the underlying discrete wave-propagation problem remains well conditioned. We give an explicit separation of these two effects for plane-wave discontinuous Galerkin (PWDG) and the ultra-weak variational formulation (UWVF), and thereby identify the condition number that belongs to the discrete operator rather than to its plane-wave coordinates. For the classical UWVF matrix, the usual one-sided scaling by the block trace Gramian is shown to preserve the generalized spectrum but not the singular-value conditioning. Symmetric trace scaling removes this coordinate distortion. On the PWDG side, the imaginary part of the Trefftz-DG form supplies a second, global graph metric; symmetric normalization in this metric produces a normal matrix and a basis-invariant condition number. The classical PWDG–UWVF equivalence then identifies the two descriptions of the same coupled transmission problem. A numerical conditioning study on an unstructured mesh, including nearly coalescing plane-wave directions, shows raw and one-sided condition numbers growing by many orders of magnitude while symmetric normalization removes most of the coordinate-driven growth. Additional wavenumber and mesh-refinement studies show that the remaining graph-normalized condition number is not a universal constant: it measures the continuity-to-inf–sup ratio of the assembled Trefftz-DG operator in its natural norm.

Keywords: Trefftz methods, plane-wave discontinuous Galerkin, ultra-weak variational formulation, basis conditioning, operator conditioning, symmetric scaling

1. Introduction

The conditioning problem addressed in this paper has been present in plane-wave Trefftz discretizations almost from the beginning of their modern development. In the ultra-weak variational formulation (UWVF) of Cessenat and Després, the unknown is an impedance trace on the element boundaries rather than a volume field, and local Trefftz solutions communicate through an interface operator [1, 2]. This formulation was attractive for high-frequency Helmholtz problems because plane waves could be used directly as local trial functions and because the global coupling was confined to element boundaries [2, 5]. Darrigrand and Monk later coupled the UWVF to a multilevel fast multipole method for exterior Maxwell problems, demonstrating the formulation in a larger-scale wave-computation setting [11]. At the same time, however, the use of

increasingly rich plane-wave sets introduced a second numerical issue that is independent of dispersion: the local impedance traces can become nearly linearly dependent [5, 17, 9].

This difficulty was already explicit in the early computational UWVF literature [5]. Huttunen, Monk and Kaipio wrote the discrete system as

$$(D - C)X = b, \tag{1}$$

where D is the block-diagonal Gram matrix of local impedance traces and C contains the interelement coupling [5]. The matrix D therefore entered the method before conditioning became a separate research topic: its element blocks were used to judge whether too many local plane waves had been retained, and the algebraic system was naturally rewritten with D^{-1} acting from the left [5, 6]. Subsequent UWVF implementations continued this practice [6, 9, 10]. In the three-dimensional Maxwell computations of Huttunen, Malinen and Monk, for example, the number of local basis functions was adjusted by constraining the condition number of the blocks D_K, and the authors explicitly noted the adverse effect of an ill-conditioned D on iterative solution [6]. Thus the classical UWVF already contained both ingredients that matter here: a local trace Gramian and a one-sided Gram scaling [5, 6, 9].

A parallel development led from UWVF to the broader family of plane-wave discontinuous Galerkin methods. Gittelson, Hiptmair and Perugia gave the h-version analysis of PWDG in 2009 and recorded the connection with UWVF within a DG framework based on averages, jumps and plane-wave local spaces [3]. This formulation made approximation, stability and numerical fluxes more transparent, and it became one of the standard Trefftz-DG models; see the later survey of Hiptmair, Moiola and Perugia [17]. Hiptmair, Moiola and Perugia subsequently extended Trefftz-DG analysis to acoustic scattering on locally refined meshes, including wave-number-explicit stability estimates and flux choices adapted to local refinement [12]. Generalized plane-wave DG constructions were later developed for variable-coefficient and anisotropic Helmholtz equations; Yuan combined a local spectral-element construction with generalized plane waves and derived approximation error estimates for that setting [13]. The equivalence between PWDG and UWVF at the standard one-half flux parameters was therefore understood well before the conditioning question studied here [3, 17]. What remained less clear was how the two formulations should be compared once their coefficient bases became badly conditioned.

By the late 2010s, plane-wave conditioning had become an explicit subject in its own right. Congreve, Gedicke and Perugia numerically quantified the conditioning of local plane-wave bases and found strong growth with the number of directions; they improved the global PWDG solve by modified Gram–Schmidt orthogonalization of the local bases [7]. More recently, Barucq, Bendali, Diaz and Tordeux developed elementwise SVD and QR strategies for the plane-wave UWVF, including truncated variants that remove numerically redundant local directions [8]. These works established that a large part of the practical ill-conditioning is local and representational: changing or reducing the local basis can improve the assembled system dramatically without changing the underlying Helmholtz transmission problem [7, 8, 10].

The conditioning question is also part of a broader finite-precision approximation literature. Perrey-Debain analyzed the finite-precision limits of plane-wave decompositions and the onset of unstable recovery as the representation becomes nearly singular [25]. Betcke and Trefethen showed in the method of particular solutions that small singular

values alone can be misleading and advocated subspace-angle information [23]; Adcock and Huybrechs later developed the general frame viewpoint in which highly redundant, ill-conditioned coordinate systems can still yield accurate approximations after regularization [22]. For Helmholtz approximation specifically, Parolin, Huybrechs and Moiola showed that stable representations can be recovered by changing the plane-wave dictionary itself, using evanescent waves to avoid exponentially large coefficients [24]. Coyle and Nigam recently studied the conditioning of DG plane-wave Trefftz matrices on a single element, including condition numbers, GMRES behavior and field error under several preconditioners [20]. Their single-element analysis isolates local matrix mechanisms; the present work addresses assembled multi-element coupling and separates local coordinate conditioning from the continuity and inf–sup properties of the global discrete operator. The UWVF convergence analysis of Buffa and Monk provides an additional bridge between the impedance-trace formulation and its DG interpretation [21].

Relation to previous work by the author.. A preliminary version of part of this work appeared as *The PWDG–UWVF Equivalence Revisited*, arXiv:2608.23822v1 [18]. That version established the direct one-half-parameter PWDG–UWVF equivalence, the impedance identity, and the paired normal forms $K + iI$ and $\frac{1}{2}I + iK_{\mathrm{U}}$. The present manuscript subsumes that preprint and extends it into a conditioning analysis. Separately, the identity $G = (A - A^*)/(2i)$, graph normalization, and trace-rank compression with a relative cutoff were used computationally in the direction-adaptive PWDG framework of [19]. The new results developed here are the separation of one-sided and symmetric UWVF trace scaling, the unitary equivalence of all exact local trace orthogonalizations, the continuity-to-inf–sup interpretation of the graph-normalized condition number, the contraction structure of the discrete UWVF transmission operator, the general positive-penalty impedance representation, and assembled multi-element studies that isolate coordinate collapse, operator conditioning, and finite-precision regularization.

The historical progression is therefore quite striking. UWVF supplied the trace Gram matrix D and the classical scaling $I - D^{-1}C$ [5, 9]; PWDG supplied a variational formulation whose imaginary part is a positive jump-and-boundary form [3, 17]; later conditioning studies supplied local orthogonalization, SVD and rank reduction [7, 8]. Yet these ingredients were usually used to improve a linear system rather than to separate two mathematically different quantities: the conditioning of the chosen plane-wave coordinate system and the conditioning of the assembled coupled operator. A large Euclidean condition number of the raw matrix mixes these effects. In a nearly dependent plane-wave basis it can be dominated by the first even when the second remains moderate.

The classical UWVF algebra contains the exact structure needed for this separation. From (1) one obtains

$$A_L := D^{-1}(D - C) = I - D^{-1}C.$$

This is the familiar one-sided formulation used throughout the computational UWVF literature [5, 6, 9, 10]. It applies the inverse trace Riesz map to the residual, and hence it has the correct generalized spectrum associated with the pair $(D - C, D)$. It does not, however, place both the trial and test coordinates in an orthonormal trace basis. The symmetric representation

$$A_T := D^{-1/2}(D - C)D^{-1/2}$$

does. The matrices A_L and A_T are similar and therefore isospectral, but their Euclidean singular values need not agree. When D is poorly conditioned, the singular-value conditioning of A_L can still contain a large coordinate contribution. This observation changes the interpretation of the classical D^{-1} scaling: it is the one-sided representation of a trace-space operator, whereas symmetric scaling exposes the singular-value conditioning in trace-orthonormal coordinates.

PWDG supplies a second, global route to the same distinction. For Trefftz fields, the imaginary part of the DG sesquilinear form is the positive jump-and-boundary norm [3, 17]. Consequently the skew-Hermitian part of the assembled PWDG matrix is itself the Gram matrix of the discrete graph norm. Symmetric normalization by this matrix produces a normal operator of the form

$$K + iI, \qquad K = K^*,$$

so its singular values are determined by a Hermitian generalized eigenproblem and are invariant under nonsingular changes of Trefftz basis. This gives an operator-conditioning quantity that is not contaminated by arbitrary scaling or near dependence of the coefficient basis.

The PWDG–UWVF equivalence, already implicit in the development of UWVF as an upwind-DG-type method and explicit in the PWDG analysis, is the structural bridge that makes the separation complete [6, 3, 17]. We use the classical equivalence at $\alpha = \beta = \delta = 1/2$ to transport the two conditioning pictures between broken-field and impedance-trace coordinates. A direct impedance-trace derivation is retained because it fixes the normalization factors and shows exactly how the UWVF trace metric and the PWDG graph metric arise from the same transmission problem. The extension to general positive PWDG penalties then gives the corresponding effective impedance and forward–reverse coupling.

The contributions of the paper are organized around this conditioning history. First, we establish that the classical UWVF scaling must be separated into its one-sided and symmetric forms, and distinguish the one-sided matrix $I - D^{-1}C$ from the symmetrically trace-scaled operator $D^{-1/2}(D - C)D^{-1/2}$, deriving direct singular-value bounds for the latter. Second, we identify the PWDG graph matrix with the skew-Hermitian part of the discrete operator and derive its normal form and basis-invariant singular values. Third, we use the PWDG–UWVF equivalence to show how these constructions represent the same coupled transmission problem in different coordinates. Finally, a numerical validation study follows the historical algebra all the way to finite precision. On a multi-element unstructured mesh we study both increasing plane-wave enrichment and deliberately coalescing directions. In the latter experiment the mesh, discrete dimension, differential equation and exact discrete solution are fixed while only the coordinate basis collapses. Raw and one-sided systems then lose physical digits despite backward-stable linear solves, whereas the symmetrically normalized formulations remain moderately conditioned and recover the field to near machine precision after the unresolved trace direction is removed.

2. PWDG in broken Trefftz fields

Let $k > 0$ be real. Throughout, $\Omega \subset \mathbb{R}^2$ is a connected polygonal domain and let $\mathcal{T}_h$ be a conforming mesh with interior edges $\mathcal{F}_h^I$ and boundary $\Gamma = \partial\Omega$. We consider

$$-\Delta u - k^2 u = 0 \quad \text{in } \Omega, \qquad \partial_n u + iku = g \quad \text{on } \Gamma. \tag{2}$$

Let V_h be a finite-dimensional broken Trefftz space, so every $v_h \in V_h$ satisfies $-\Delta v_h - k^2 v_h = 0$ on each element; this is the standard Trefftz-DG setting for Helmholtz discretizations [3, 17].

For an interior edge $e = \partial K \cap \partial K'$, with outward normals n_K and $n_{K'}$, define the actual DG jumps

$$[\![v]\!]_N = v_K n_K + v_{K'} n_{K'}, \qquad [\![\nabla v]\!]_N = \partial_{n_K} v_K + \partial_{n_{K'}} v_{K'}, \tag{3}$$

and use the arithmetic averages $\{\{v\}\}$ and $\{\{\nabla v\}\}$.

The primal Trefftz-DG formulation [3, 17] is: find $u_h \in V_h$ such that

$$A_h(u_h, v_h) = \ell_h(v_h) \qquad \forall v_h \in V_h, \tag{4}$$

where, for $\alpha, \beta > 0$ and $0 < \delta < 1$,

$$\begin{aligned} A_h(u,v) = \int_{\mathcal{F}_h^I} \Big(& \{\{u\}\}\, [\![\nabla \bar{v}]\!]_N - \{\{\nabla u\}\} \cdot [\![\bar{v}]\!]_N + \alpha i k\, [\![u]\!]_N \cdot [\![\bar{v}]\!]_N \\ & - \beta (ik)^{-1} [\![\nabla u]\!]_N [\![\nabla \bar{v}]\!]_N \Big)\, ds \\ + \int_\Gamma \Big(& (1-\delta) i k u \bar{v} + (1-\delta) u\, \overline{\partial_n v} - \delta\, \partial_n u\, \bar{v} - \delta (ik)^{-1} \partial_n u\, \overline{\partial_n v} \Big)\, ds, \end{aligned} \tag{5}$$

and

$$\ell_h(v) = \int_\Gamma g \Big((1-\delta)\bar{v} - \delta (ik)^{-1} \overline{\partial_n v} \Big)\, ds. \tag{6}$$

The associated Trefftz discontinuous Galerkin (TDG) norm is

$$\begin{aligned} \|v\|_{\mathrm{TDG}}^2 = {} & k^{-1} \|\beta^{1/2} [\![\nabla v]\!]_N\|_{\mathcal{F}_h^I}^2 + k \|\alpha^{1/2} [\![v]\!]_N\|_{\mathcal{F}_h^I}^2 \\ & + k^{-1} \|\delta^{1/2} \partial_n v\|_\Gamma^2 + k \|(1-\delta)^{1/2} v\|_\Gamma^2. \end{aligned} \tag{7}$$

3. PWDG written directly as a UWVF

For each element K, define the outgoing and incoming impedance variables

$$x_K(v) := -\partial_{n_K} v + ikv, \qquad y_K(v) := \partial_{n_K} v + ikv. \tag{8}$$

Following Cessenat and Després [1, 2], $x_K(v)$ is the outgoing impedance and $y_K(v)$ is the incoming impedance. The outgoing impedance trace is the basic UWVF unknown, and its local injectivity is part of the well-posed local trace-to-field construction underlying UWVF [1, 2, 8]. Indeed, if $x_K(v) = 0$, then $\partial_{n_K} v = ikv$ on ∂K, and Green's identity gives

$$\int_K \left(|\nabla v|^2 - k^2 |v|^2 \right) dx = ik \int_{\partial K} |v|^2\, ds.$$

The left-hand side is real, so $v = 0$ on ∂K; then also $\partial_{n_K} v = 0$, and uniqueness for the Cauchy problem implies $v \equiv 0$ in K. Consequently the local scattering operator is well defined on the range of the outgoing trace and satisfies

$$y_K(v) = F_K x_K(v). \tag{9}$$

Thus the discrete UWVF trace space is precisely

$$\mathcal{X}_h^- := \{(x_K(v_h|_K))_{K\in\mathcal{T}_h} : v_h \in V_h\}.$$

If $\{\Phi_j\}$ is a basis of V_h, we use $\{(x_K(\Phi_j|_K))_K\}$ as the corresponding UWVF trace basis. With this identification, PWDG and UWVF matrices may be compared in the same coefficient vector without any hidden change of basis.

For $u, v \in V_h$, define the pulled-back standard UWVF form

$$\begin{aligned}\mathcal{U}_h(u,v) = &\sum_{K\in\mathcal{T}_h} \int_{\partial K} x_K(u_K)\,\overline{x_K(v_K)}\,ds \\ &- \sum_{e=\partial K\cap\partial K'\in\mathcal{F}_h^I} \int_e \Big(x_{K'}(u_{K'})\,\overline{y_K(v_K)} + x_K(u_K)\,\overline{y_{K'}(v_{K'})}\Big)\,ds, \end{aligned} \tag{10}$$

and

$$\mathcal{L}_h(v) = \sum_{K\in\mathcal{T}_h} \int_{\partial K\cap\Gamma} g\,\overline{y_K(v_K)}\,ds. \tag{11}$$

Equation (10) is exactly the usual UWVF with outgoing trace unknown x_K and incoming trace $y_K = F_K x_K$, pulled back through (8) to broken Trefftz fields [1, 5, 3].

Theorem 1 (PWDG is the pulled-back UWVF)**.** *At*

$$\alpha = \beta = \delta = \frac{1}{2},$$

the PWDG formulation (4) *is precisely*

$$\frac{i}{2k}\,\mathcal{U}_h(u_h, v_h) = \frac{i}{2k}\,\mathcal{L}_h(v_h) \qquad \forall v_h \in V_h. \tag{12}$$

Equivalently,

$$A_h(u,v) = \frac{i}{2k}\mathcal{U}_h(u,v), \qquad \ell_h(v) = \frac{i}{2k}\mathcal{L}_h(v). \tag{13}$$

Thus the PWDG broken-field equation is a UWVF written in broken Trefftz coordinates.

Proof. Consider first one interior edge $e = \partial K \cap \partial K'$. The contribution of this edge to $\mathcal{U}_h$ is

$$\begin{aligned}\mathcal{U}_e(u,v) = &\,\langle x_K(u_K), x_K(v_K)\rangle_e + \langle x_{K'}(u_{K'}), x_{K'}(v_{K'})\rangle_e \\ &- \langle x_{K'}(u_{K'}), y_K(v_K)\rangle_e - \langle x_K(u_K), y_{K'}(v_{K'})\rangle_e\,. \end{aligned} \tag{14}$$

For example,

$$x_K(u_K)\,\overline{x_K(v_K)} = \partial_{n_K} u_K\,\overline{\partial_{n_K} v_K} + ik\,\partial_{n_K} u_K\,\bar v_K - ik\,u_K\,\overline{\partial_{n_K} v_K} + k^2 u_K \bar v_K,$$

and the three remaining products have the same four value–flux components with the corresponding orientation signs. Collecting the value–value and flux–flux terms, then the two mixed terms, and using $n_{K'} = -n_K$ gives

$$\begin{aligned}\frac{i}{2k}\mathcal{U}_e(u,v) = &\int_e \Big(\{\{u\}\}\,[\![\nabla\bar v]\!]_N - \{\{\nabla u\}\}\cdot[\![\bar v]\!]_N\Big)\,ds \\ &+ \frac{ik}{2}\,\langle [\![u]\!]_N, [\![v]\!]_N\rangle_e + \frac{i}{2k}\,\langle [\![\nabla u]\!]_N, [\![\nabla v]\!]_N\rangle_e\,. \end{aligned} \tag{15}$$

This is exactly the interior PWDG edge form with $\alpha = \beta = 1/2$.

On a physical boundary edge, the diagonal UWVF term gives

$$\frac{i}{2k}\,\langle x(u), x(v)\rangle_\Gamma = \int_\Gamma \Big(\frac{ik}{2}u\bar{v} + \frac{1}{2}u\,\overline{\partial_n v} - \frac{1}{2}\partial_n u\,\bar{v} + \frac{i}{2k}\partial_n u\,\overline{\partial_n v}\Big)\,ds, \tag{16}$$

which is the boundary part of (5) at $\delta = 1/2$. Finally,

$$\frac{i}{2k}\int_\Gamma g\,\overline{y(v)}\,ds = \int_\Gamma g\Big(\frac{1}{2}\bar{v} + \frac{i}{2k}\overline{\partial_n v}\Big)\,ds = \ell_h(v), \tag{17}$$

again because $-(ik)^{-1} = i/k$. Summing over all edges proves (13). □

Remark 1 (Where the scalar factor comes from)**.** *The factor $i/(2k)$ is the normalization relating the two methods. If the impedance variables are rescaled by $(2k)^{-1/2}$, then the pulled-back UWVF matrix is simply $-iA_{\mathrm{P}}$. If the standard unscaled impedance variables* (8) *are retained, then the matrix relation is instead*

$$A_{\mathrm{U}}^{\mathrm{std}} = -2ik\,A_{\mathrm{P}}. \tag{18}$$

The two versions have identical condition numbers and identical normalized spectra.

4. The impedance identity behind the PWDG jumps

The preceding theorem uses the physical impedance and the one-half parameters appearing in the classical PWDG–UWVF correspondence [3, 17]. We now keep $\alpha, \beta > 0$ arbitrary and show directly how the PWDG jumps are encoded by two oriented impedance transmission conditions.

Set

$$\eta_{\alpha,\beta} = k\sqrt{\alpha/\beta}, \qquad \rho_{\alpha,\beta} = 2\sqrt{\alpha\beta}. \tag{19}$$

Define the corresponding weighted impedance variables

$$x_K^{\alpha,\beta}(v) := -\sqrt{\frac{\beta}{k}}\,\partial_{n_K} v + i\sqrt{\alpha k}\,v, \qquad y_K^{\alpha,\beta}(v) := \sqrt{\frac{\beta}{k}}\,\partial_{n_K} v + i\sqrt{\alpha k}\,v. \tag{20}$$

Equivalently,

$$x_K^{\alpha,\beta}(v) = \sqrt{\beta/k}\,(-\partial_{n_K} + i\eta_{\alpha,\beta})v, \qquad y_K^{\alpha,\beta}(v) = \sqrt{\beta/k}\,(\partial_{n_K} + i\eta_{\alpha,\beta})v.$$

Proposition 1 (Impedance variables and DG jumps)**.** *Let $e = \partial K \cap \partial K'$ be an interior edge. Then*

$$\begin{aligned}
&\frac{1}{2}\Big\langle x_K^{\alpha,\beta}(u_K) - y_{K'}^{\alpha,\beta}(u_{K'}), x_K^{\alpha,\beta}(v_K) - y_{K'}^{\alpha,\beta}(v_{K'})\Big\rangle_e \\
&\quad + \frac{1}{2}\Big\langle x_{K'}^{\alpha,\beta}(u_{K'}) - y_K^{\alpha,\beta}(u_K), x_{K'}^{\alpha,\beta}(v_{K'}) - y_K^{\alpha,\beta}(v_K)\Big\rangle_e \\
&\qquad = \alpha k\,\langle [\![u]\!]_N, [\![v]\!]_N\rangle_e + \frac{\beta}{k}\,\langle [\![\nabla u]\!]_N, [\![\nabla v]\!]_N\rangle_e\,.
\end{aligned} \tag{21}$$

Thus the two oriented impedance mismatches on the left are exactly the complex packaging of the Dirichlet and normal-flux jumps on the right.

Proof. Because $n_{K'} = -n_K$,

$$x_K^{\alpha,\beta}(u_K) - y_{K'}^{\alpha,\beta}(u_{K'}) = -\sqrt{\frac{\beta}{k}}\,[\![\nabla u]\!]_N + i\sqrt{\alpha k}\,([\![u]\!]_N \cdot n_K), \tag{22}$$

$$x_{K'}^{\alpha,\beta}(u_{K'}) - y_K^{\alpha,\beta}(u_K) = -\sqrt{\frac{\beta}{k}}\,[\![\nabla u]\!]_N - i\sqrt{\alpha k}\,([\![u]\!]_N \cdot n_K). \tag{23}$$

The same identities hold for v. Put

$$a_u = -\sqrt{\beta/k}\,[\![\nabla u]\!]_N, \qquad b_u = \sqrt{\alpha k}\,([\![u]\!]_N \cdot n_K),$$

and define a_v, b_v analogously. The two mismatches are $a_u + ib_u$ and $a_u - ib_u$, so

$$\frac{1}{2}\langle a_u + ib_u, a_v + ib_v\rangle_e + \frac{1}{2}\langle a_u - ib_u, a_v - ib_v\rangle_e = \langle a_u, a_v\rangle_e + \langle b_u, b_v\rangle_e.$$

The mixed value–flux terms therefore cancel exactly. Since $[\![u]\!]_N$ and $[\![v]\!]_N$ are parallel to n_K on e, the two remaining products are precisely those on the right-hand side of (21). □

At $\alpha = \beta = 1/2$, the weighted variables are exactly $(2k)^{-1/2}x_K$ and $(2k)^{-1/2}y_K$. Proposition 1 therefore explains the two interior one-half factors in the classical equivalence: the Dirichlet and normal-flux jump terms are the two-channel real representation of the two oriented UWVF impedance transmission conditions.

5. General positive PWDG penalties as weighted UWVF couplings

Proposition 1 identifies the positive jump form generated by the weighted impedance variables. The flux parameters and Trefftz structure are those of the standard PWDG formulation [3, 17]. The complete PWDG edge operator has an equally direct impedance representation.

For an interior edge define the consistency and jump forms

$$C_e(u,v) = \int_e \Big(\{\{u\}\}\,[\![\nabla \bar{v}]\!]_N - \{\{\nabla u\}\}\cdot[\![\bar{v}]\!]_N\Big)\,ds, \tag{24}$$

$$G_e^{\alpha,\beta}(u,v) = \alpha k\,\langle [\![u]\!]_N, [\![v]\!]_N\rangle_e + \frac{\beta}{k}\,\langle [\![\nabla u]\!]_N, [\![\nabla v]\!]_N\rangle_e\,. \tag{25}$$

Then

$$A_{\mathrm{P},e}(u,v) = C_e(u,v) + iG_e^{\alpha,\beta}(u,v). \tag{26}$$

Using the weighted impedance variables (20), define the one-way pulled-back UWVF edge form by

$$\begin{aligned}\mathcal{U}_e^{\rightarrow}(u,v) = &\left\langle x_K^{\alpha,\beta}(u_K), x_K^{\alpha,\beta}(v_K)\right\rangle_e + \left\langle x_{K'}^{\alpha,\beta}(u_{K'}), x_{K'}^{\alpha,\beta}(v_{K'})\right\rangle_e \\ &- \left\langle x_{K'}^{\alpha,\beta}(u_{K'}), y_K^{\alpha,\beta}(v_K)\right\rangle_e - \left\langle x_K^{\alpha,\beta}(u_K), y_{K'}^{\alpha,\beta}(v_{K'})\right\rangle_e.\end{aligned} \tag{27}$$

A direct expansion gives

$$\mathcal{U}_e^{\rightarrow} = G_e^{\alpha,\beta} - i\rho_{\alpha,\beta}C_e. \tag{28}$$

Because $\rho_{\alpha,\beta} = 2\sqrt{\alpha\beta}$, the condition $\rho_{\alpha,\beta} = 1$ is equivalent to $4\alpha\beta = 1$. Consequently, on this curve

$$A_{\mathrm{P},e} = i\,\mathcal{U}_e^{\rightarrow}. \tag{29}$$

Thus the complete PWDG interior edge form is an ordinary one-way UWVF on the entire curve $4\alpha\beta = 1$. The physical-impedance condition $\eta_{\alpha,\beta} = k$ intersects this curve only at $\alpha = \beta = 1/2$.

For arbitrary positive α, β, introduce also the reverse-orientation one-way form

$$\begin{aligned}\mathcal{U}_e^{\leftarrow}(u,v) = &\left\langle y_K^{\alpha,\beta}(u_K), y_K^{\alpha,\beta}(v_K)\right\rangle_e + \left\langle y_{K'}^{\alpha,\beta}(u_{K'}), y_{K'}^{\alpha,\beta}(v_{K'})\right\rangle_e \\ &- \left\langle y_{K'}^{\alpha,\beta}(u_{K'}), x_K^{\alpha,\beta}(v_K)\right\rangle_e - \left\langle y_K^{\alpha,\beta}(u_K), x_{K'}^{\alpha,\beta}(v_{K'})\right\rangle_e .\end{aligned} \tag{30}$$

It is obtained from the forward form by reversing the sign of the normal-derivative channel in the impedance variables. Therefore the same expansion that gives (28) yields

$$\mathcal{U}_e^{\leftarrow} = G_e^{\alpha,\beta} + i\rho_{\alpha,\beta} C_e. \tag{31}$$

This operation reverses the orientation. It is distinct from the Hilbert-space adjoint of $\mathcal{U}_e^{\rightarrow}$, since edgewise C_e is generally non-Hermitian.

Set $\rho = 2\sqrt{\alpha\beta}$ and define the two-orientation form

$$\mathcal{U}_e^{\alpha,\beta} = \frac{\rho+1}{2\rho}\mathcal{U}_e^{\rightarrow} + \frac{\rho-1}{2\rho}\mathcal{U}_e^{\leftarrow}. \tag{32}$$

Using (28) and (31),

$$\mathcal{U}_e^{\alpha,\beta} = G_e^{\alpha,\beta} - iC_e, \qquad A_{\mathrm{P},e} = i\mathcal{U}_e^{\alpha,\beta}. \tag{33}$$

No extra tuning parameter has been introduced: both coefficients in (32) are fixed by the original PWDG penalties. They sum to one, so (32) is an affine combination. It is a convex combination only when $\rho \geq 1$; for $\rho < 1$ the coefficient of the reverse orientation is negative.

The effective impedance $\eta_{\alpha,\beta} = k\sqrt{\alpha/\beta}$ is therefore a genuine design parameter for the local transmission variables, while $\rho = 2\sqrt{\alpha\beta}$ controls their forward–reverse weighting. Section 8 tests the resulting graph conditioning across both equal-penalty choices and the one-way curve $4\alpha\beta = 1$. A smaller graph condition number by itself measures only the discrete algebra; changing penalties also changes the discrete variational problem and must be assessed together with approximation error. We therefore use the parameter sweep only to test the structural persistence of the graph normalization, not to recommend an "optimal" penalty.

The boundary parameter δ has the analogous Riesz interpretation

$$\frac{\delta}{k}\|\partial_n v\|_\Gamma^2 + (1-\delta)k\|v\|_\Gamma^2,$$

but the standard physical one-way UWVF boundary term occurs at $\delta = 1/2$, exactly as shown in Theorem 1. For the classical PWDG–UWVF equivalence, therefore, all three parameters are one-half.

6. Graph normalization and operator conditioning

Let $\{\Phi_j\}_{j=1}^N$ be a basis of V_h. The underlying TDG stability identity is classical in the PWDG analysis [3, 17]; here it is used as a conditioning identity at matrix level. Define the PWDG matrix A_{P} and the TDG Gram matrix G_{P} by

$$(A_{\mathrm{P}})_{j\ell} = A_h(\Phi_\ell, \Phi_j), \qquad (G_{\mathrm{P}})_{j\ell} = (\Phi_\ell, \Phi_j)_{\mathrm{TDG}}.$$

Thus, for $v = \sum_j c_j \Phi_j$,

$$c^* A_{\mathrm{P}} c = A_h(v, v), \qquad c^* G_{\mathrm{P}} c = \|v\|_{\mathrm{TDG}}^2.$$

Theorem 2 (PWDG Riesz normal form)**.** *The TDG Gram matrix G_{P} is positive definite on V_h. Moreover,*

$$G_{\mathrm{P}} = \frac{A_{\mathrm{P}} - A_{\mathrm{P}}^*}{2i}. \tag{34}$$

With $B_{\mathrm{P}} = G_{\mathrm{P}}^{-1/2}$,

$$B_{\mathrm{P}}^* A_{\mathrm{P}} B_{\mathrm{P}} = K + iI, \qquad K = K^*. \tag{35}$$

Hence the Riesz-normalized PWDG matrix is normal and its spectrum lies on $\operatorname{Im} z = 1$.

Proof. If $\|v\|_{\mathrm{TDG}} = 0$, then both interior jumps vanish, so the broken Trefftz field and its normal derivative match across every interior edge. Hence v is a global Helmholtz solution. The two positive boundary terms in (7) give $v = 0$ and $\partial_n v = 0$ on Γ. Uniqueness for the Helmholtz Cauchy problem on a connected domain then yields $v \equiv 0$. Thus $\|\cdot\|_{\mathrm{TDG}}$ is a norm on V_h, and its Gram matrix is positive definite in exact arithmetic.

We next derive the Trefftz-DG stability identity directly. On an interior edge $e = \partial K^+ \cap \partial K^-$, the consistency contribution is

$$C_e(v, v) = \int_e \Big(\{\{v\}\} \, [\![\nabla \bar{v}]\!]_N - \{\{\nabla v\}\} \cdot [\![\bar{v}]\!]_N \Big) \, ds.$$

Expanding the averages and jumps gives

$$\begin{aligned} 2C_e(v, v) = &\int_e \Big(v^+ \overline{\partial_{n^+} v^+} - \partial_{n^+} v^+ \, \bar{v}^+ + v^- \overline{\partial_{n^-} v^-} - \partial_{n^-} v^- \, \bar{v}^- \Big) \, ds \\ &+ 2 \operatorname{Re} \int_e \Big(v^+ \overline{\partial_{n^-} v^-} + v^- \overline{\partial_{n^+} v^+} \Big) \, ds. \end{aligned}$$

The second line is real, and therefore

$$\operatorname{Im} C_e(v, v) = \operatorname{Im} \int_e \Big(v^+ \overline{\partial_{n^+} v^+} + v^- \overline{\partial_{n^-} v^-} \Big) \, ds. \tag{36}$$

On the physical boundary, the consistency part of (5) has imaginary part $\operatorname{Im}(v \overline{\partial_n v})$. Since v is Trefftz on every element, Green's identity gives

$$\sum_{K \in \mathcal{T}_h} \int_{\partial K} v \, \overline{\partial_{n_K} v} \, ds = \sum_{K \in \mathcal{T}_h} \int_K \left(|\nabla v|^2 - k^2 |v|^2 \right) dx \in \mathbb{R}.$$

Splitting the element boundaries into interior and physical edges and using (36), the imaginary parts of all consistency terms cancel. The remaining imaginary terms are exactly

$$\begin{aligned}\operatorname{Im} A_h(v,v) = \alpha k\|[\![v]\!]_N\|^2_{\mathcal{F}_h^I} + \frac{\beta}{k}\|[\![\nabla v]\!]_N\|^2_{\mathcal{F}_h^I} \\ + (1-\delta)k\|v\|^2_\Gamma + \frac{\delta}{k}\|\partial_n v\|^2_\Gamma = \|v\|^2_{\mathrm{TDG}}.\end{aligned}$$

For every coefficient vector c, this says

$$c^*\left(\frac{A_{\mathrm{P}} - A_{\mathrm{P}}^*}{2i} - G_{\mathrm{P}}\right)c = 0.$$

The matrix in parentheses is Hermitian. A Hermitian matrix whose quadratic form vanishes identically has only zero eigenvalues, so it is the zero matrix. Hence (34) follows. Writing $H = (A_{\mathrm{P}} + A_{\mathrm{P}}^*)/2$, one has $A_{\mathrm{P}} = H + iG_{\mathrm{P}}$, hence

$$B_{\mathrm{P}}^* A_{\mathrm{P}} B_{\mathrm{P}} = B_{\mathrm{P}}^* H B_{\mathrm{P}} + iI,$$

and $K = B_{\mathrm{P}}^* H B_{\mathrm{P}}$ is Hermitian. □

If $Hx = \mu G_{\mathrm{P}} x$, then μ is invariant under every nonsingular change of Trefftz basis. This basis-robust viewpoint is closely related to the function-space philosophy of operator preconditioning [4]. The normalized eigenvalues are $\mu + i$, and because (35) is normal its singular values are

$$\sigma_j = \sqrt{1+\mu_j^2}. \tag{37}$$

Consequently

$$\kappa_2(\widehat{A}_{\mathrm{P}}) = \frac{\max_j \sqrt{1+\mu_j^2}}{\min_j \sqrt{1+\mu_j^2}}. \tag{38}$$

The denominator is governed by the generalized eigenvalue of smallest absolute value, while the numerator is governed by the largest absolute generalized eigenvalue.

Proposition 2 (Continuity and inf–sup interpretation)**.** *Let* $\widehat{A}_{\mathrm{P}} = G_{\mathrm{P}}^{-1/2} A_{\mathrm{P}} G_{\mathrm{P}}^{-1/2}$. *Define*

$$C_h := \sup_{0\neq u,v\in V_h} \frac{|A_h(u,v)|}{\|u\|_{\mathrm{TDG}}\,\|v\|_{\mathrm{TDG}}}, \qquad \gamma_h := \inf_{0\neq u\in V_h}\sup_{0\neq v\in V_h} \frac{|A_h(u,v)|}{\|u\|_{\mathrm{TDG}}\,\|v\|_{\mathrm{TDG}}}.$$

Then

$$\|\widehat{A}_{\mathrm{P}}\|_2 = C_h, \qquad \sigma_{\min}(\widehat{A}_{\mathrm{P}}) = \gamma_h, \qquad \kappa_2(\widehat{A}_{\mathrm{P}}) = \frac{C_h}{\gamma_h}. \tag{39}$$

Moreover $\gamma_h \geq 1$*, and therefore*

$$\kappa_2(\widehat{A}_{\mathrm{P}}) \leq C_h. \tag{40}$$

Thus the graph-normalized condition number is the discrete continuity-to-inf–sup ratio of the PWDG form in the TDG norm; it is a finite-dimensional counterpart of the quasi-optimality ratio appearing in the Trefftz-DG stability analysis [3, 17].

Proof. For coefficient vectors x, y, set $u = \sum_j (G_{\mathrm{P}}^{-1/2}x)_j \Phi_j$ and $v = \sum_j (G_{\mathrm{P}}^{-1/2}y)_j \Phi_j$. Then $\|u\|_{\mathrm{TDG}} = \|x\|_2$, $\|v\|_{\mathrm{TDG}} = \|y\|_2$, and $y^* \widehat{A}_{\mathrm{P}} x = A_h(u, v)$. The first two identities in (39) are therefore exactly the operator norm and smallest singular value of $\widehat{A}_{\mathrm{P}}$. The ratio gives the condition number. Finally, Theorem 2 gives $\widehat{A}_{\mathrm{P}} = K + iI$ with $K = K^*$, so every singular value is $\sqrt{1+\mu_j^2} \geq 1$. Hence $\gamma_h \geq 1$. □

This proposition gives the graph condition number a direct variational meaning. It also explains why it need not stay constant under enrichment, mesh refinement, or changes of wavenumber: those changes alter the continuity constant and the discrete inf–sup constant even after all coordinate distortion has been removed. The pencil (H, G_{P}) therefore describes the operator spectrum in the TDG norm, whereas $\kappa_2(A_{\mathrm{P}})$ additionally contains basis conditioning. If tiny eigenvalues of G_{P} are truncated, the same normal form and interpretation hold on the retained positive-definite range.

We now specialize the UWVF matrix relations to the classical case $\alpha = \beta = \delta = 1/2$, which is the only case used in the numerical experiment. Let A_{U}^w denote the one-way UWVF matrix in the weighted trace basis induced by $x_K^{1/2,1/2} = (2k)^{-1/2} x_K$. By (29) and the trace-space identification of Section 3,

$$A_{\mathrm{U}}^w = -iA_{\mathrm{P}}, \qquad G_{\mathrm{U}}^w := A_{\mathrm{U}}^w + (A_{\mathrm{U}}^w)^* = 2G_{\mathrm{P}}. \tag{41}$$

Hence, with $B_{\mathrm{U}}^w = (G_{\mathrm{U}}^w)^{-1/2}$,

$$(B_{\mathrm{U}}^w)^* A_{\mathrm{U}}^w B_{\mathrm{U}}^w = \frac{1}{2} I + iK_{\mathrm{U}}, \qquad K_{\mathrm{U}} = K_{\mathrm{U}}^*, \qquad K_{\mathrm{U}} = -\frac{1}{2} K. \tag{42}$$

Thus the two normalized matrices are not independent geometries: the UWVF normal form is exactly $-i/2$ times the PWDG normal form.

At the standard one-half parameters, the physical impedance variables (8) are $\sqrt{2k}$ times the weighted traces. Therefore

$$A_{\mathrm{U}}^{\mathrm{std}} = -2ikA_{\mathrm{P}}, \qquad G_{\mathrm{U}}^{\mathrm{std}} := A_{\mathrm{U}}^{\mathrm{std}} + (A_{\mathrm{U}}^{\mathrm{std}})^* = 4kG_{\mathrm{P}}. \tag{43}$$

After Riesz normalization, the factor $2k$ disappears. The standard UWVF therefore has the same normalized form (42), separating a trace-normalization convention from the intrinsic operator geometry.

7. Classical UWVF scaling and conditioning

The original discrete UWVF literature writes the algebraic problem in the form

$$(D - C)X = b, \qquad (I - D^{-1}C)X = D^{-1}b, \tag{44}$$

where $D = \mathrm{blockdiag}(D_K)$ with

$$(D_K)_{j\ell} = \int_{\partial K} x_K(\Phi_\ell)\, \overline{x_K(\Phi_j)}\, ds,$$

so that $D = D^* > 0$ is block diagonal, while C collects the off-element transmission terms; for the unit-impedance boundary condition used here there is no additional left-hand-side

boundary exchange term; see [5, 6, 9]. The explicit left-scaled form $I - D^{-1}C$ is also used in later UWVF studies of alternative local bases [9, 10]. The second form is usually regarded as a natural block left-preconditioning because D^{-1} is local [5, 6, 9]. The UWVF literature also recognized that large condition numbers of the element blocks of D are a principal source of numerical instability and used basis changes, alternative local bases, and later explicit rank reduction to control them [5, 6, 9, 10, 8].

There are, however, two distinct matrix representations hidden in (44). Define

$$A_L := I - D^{-1}C, \qquad A_T := D^{-1/2}(D - C)D^{-1/2} = I - D^{-1/2}CD^{-1/2}. \tag{45}$$

Then

$$A_L = D^{-1/2}A_T D^{1/2}. \tag{46}$$

Thus A_L and A_T are similar and have exactly the same eigenvalues, but generally not the same Euclidean singular values. The distinction is important because A_T, rather than A_L, is the representation of the UWVF coupling in coordinates orthonormal with respect to the trace Gramian D. Indeed, for $\widehat{X} = D^{1/2}X$,

$$X^*DX = \|\widehat{X}\|_2^2,$$

and the action of $D^{-1}C$ in the D-norm is represented in Euclidean coordinates by $D^{-1/2}CD^{-1/2}$.

Proposition 3 (One-sided versus symmetric trace scaling)**.** *Let $D = D^* > 0$, $A_U = D - C$, and define A_L, A_T by* (45)*. Then*

$$\kappa_2(A_L) \le \kappa_2(D)\,\kappa_2(A_T), \qquad \kappa_2(A_T) \le \kappa_2(D)\,\kappa_2(A_L). \tag{47}$$

Proof. Equation (46) and submultiplicativity give

$$\|A_L\|_2 \le \|D^{-1/2}\|_2\,\|A_T\|_2\,\|D^{1/2}\|_2,$$

Applying the same argument to

$$A_L^{-1} = D^{-1/2}A_T^{-1}D^{1/2}$$

gives

$$\|A_L^{-1}\|_2 \le \|D^{-1/2}\|_2\,\|A_T^{-1}\|_2\,\|D^{1/2}\|_2.$$

Multiplying the estimates for A_L and A_L^{-1} yields the first bound in (47); applying the same argument to the reverse similarity yields the second. □

Proposition 4 (Canonical singular values of local trace orthogonalization)**.** *Let $D = LL^*$ be any nonsingular factorization of the trace Gramian, and set*

$$A_L^{\mathrm{sym}} := L^{-1}(D - C)L^{-*}.$$

Then A_L^{sym} is unitarily similar to A_T. In particular,

$$\sigma_j(A_L^{\mathrm{sym}}) = \sigma_j(A_T), \qquad \kappa_2(A_L^{\mathrm{sym}}) = \kappa_2(A_T). \tag{48}$$

Thus exact local Cholesky, QR, modified Gram–Schmidt, and symmetric-square-root orthogonalizations all expose the same trace-normalized singular values; truncated SVD or QR realizes the corresponding statement on the retained numerical range.

Proof. Define $W = D^{1/2}L^{-*}$. Since $D = LL^*$,

$$W^*W = L^{-1}DL^{-*} = I,$$

so W is unitary. Direct multiplication gives

$$W^*A_TW = L^{-1}(D - C)L^{-*} = A_L^{\mathrm{sym}}.$$

Unitary similarity preserves singular values and the Euclidean condition number. □

Proposition 4 places earlier local orthogonalization results in the present framework. The modified Gram–Schmidt strategy of Congreve, Gedicke and Perugia and the local QR/SVD strategies of Barucq, Bendali, Diaz and Tordeux already realize the singular values of the canonical trace-normalized operator, exactly when no modes are dropped and on a retained numerical range when truncation is used [7, 8]. These apparently different local orthogonalizations are therefore representations of the same trace-space operator and can be cleanly separated from the global graph normalization. Proposition 4 makes a separate numerical comparison of exact modified Gram–Schmidt, QR and Cholesky unnecessary: in exact arithmetic they produce identical singular values after trace orthogonalization.

Proposition 5 (Local and global graph metrics)**.** *For the unit-impedance UWVF matrix $A_U = D - C$ associated with* (2)*, the physical-boundary impedance data are absorbed into the right-hand side and C has only off-element blocks. Therefore*

$$\mathrm{blockdiag}(A_U + A_U^*) = 2D. \tag{49}$$

Thus the trace Gramian is exactly one half of the element-block diagonal of the global UWVF graph Gramian. Local trace normalization and global graph normalization are the same Riesz construction applied at two different coupling scales.

Proof. The diagonal element blocks of A_U are those of D, while C contains only interelement couplings for the unit-impedance boundary condition in (2). Hence the diagonal element blocks of $A_U + A_U^*$ equal $2D_K$ element by element, which gives (49). □

Remark 2. *Equation* (49) *identifies the local UWVF trace metric as the element-block restriction of the global graph metric. It provides the precise algebraic link between inexpensive local trace normalization and the assembled graph normalization.*

Proposition 6 (Discrete UWVF contraction)**.** *For the unit-impedance UWVF considered here, let*

$$U_D := D^{-1/2}CD^{-1/2}, \qquad q_h := \|U_D\|_2.$$

Then

$$q_h \le 1. \tag{50}$$

Whenever the discrete transmission space satisfies the strict contraction condition $q_h < 1$,

$$1 - q_h \le \sigma_j(A_T) \le 1 + q_h, \qquad \kappa_2(A_T) \le \frac{1 + q_h}{1 - q_h}. \tag{51}$$

The strict inequality is a discrete-space property of the projected transmission map; it is not inferred here from uniqueness of the continuous boundary-value problem.

Proof. For a Trefftz field on one element, the definitions of the outgoing and incoming traces give

$$\begin{aligned}\int_{\partial K} \left(|y_K(v)|^2 - |x_K(v)|^2\right) ds &= 4k \ \mathrm{Im} \int_{\partial K} v\, \overline{\partial_{n_K} v}\, ds \\ &= 4k \ \mathrm{Im} \int_K \left(|\nabla v|^2 - k^2 |v|^2\right) dx = 0,\end{aligned}$$

where the second equality is Green's identity and the last integral is real. Hence

$$\int_{\partial K} |y_K(v)|^2 \, ds = \int_{\partial K} |x_K(v)|^2 \, ds,$$

so the local impedance-to-impedance map $F_K : x_K(v) \mapsto y_K(v)$ is an isometry on the discrete trace range; this is the energy identity underlying the UWVF construction [1, 2]. Let $F = \mathrm{blockdiag}(F_K)$ and let F^* denote its adjoint in the trace inner product; since F is an isometry on the discrete trace range, so is F^*. Let Π_I be the operator that swaps the two oriented traces across every interior edge and sets the physical-boundary component to zero. Thus $\|\Pi_I\| \le 1$. Finally let P_h be the trace-inner-product orthogonal projection onto the discrete outgoing trace space X_h^-. Reading the trial–test pairing in (10) shows that the coupling represented by U_D in D-orthonormal coordinates is the composition

$$U_D \cong P_h F^* \Pi_I.$$

Each factor has norm at most one, and therefore $\|U_D\|_2 \le 1$.

Finally $A_T = I - U_D$. If the computed discrete space satisfies $q_h < 1$, the triangle and reverse-triangle inequalities give (51). Strict contraction is a property of this projected discrete transmission map and must be checked from the discrete space or established under an additional invariance-exclusion hypothesis. □

The trace normalization A_T, which orthonormalizes the local trace metric in the same spirit as earlier UWVF basis orthogonalization and SVD/QR strategies [5, 7, 8], should not be confused with the graph–Riesz normalization $(G_{\mathrm{U}}^{\mathrm{std}})^{-1/2} A_{\mathrm{U}}^{\mathrm{std}} (G_{\mathrm{U}}^{\mathrm{std}})^{-1/2}$ derived above. The matrix D measures the norm of the UWVF trace unknown itself, whereas $G_{\mathrm{U}}^{\mathrm{std}} = A_{\mathrm{U}}^{\mathrm{std}} + (A_{\mathrm{U}}^{\mathrm{std}})^*$ measures the global transmission-defect norm induced by the Trefftz-DG stability identity. The former removes coordinate distortion already at the level of the trace basis; the latter additionally exposes the intrinsic graph geometry of the coupled discrete problem. Section 8 compares all three levels numerically.

8. Numerical validation

We use the sixteen-triangle unstructured triangulation of the unit square shown in Figure 1, with $k = 4$ and p equispaced centroided plane waves per element, following the standard plane-wave Trefftz practice of using angularly distributed local directions [3, 17, 7]. The same mesh is used in every numerical experiment below.

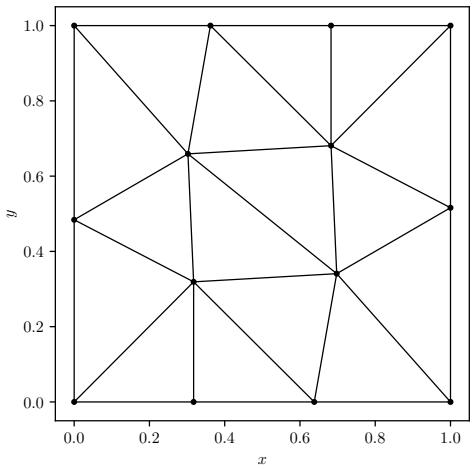


Figure 1: Sixteen-triangle unstructured mesh used in all numerical experiments.

No numerical quadrature is used. Closed-form edge integration is a standard computational advantage of plane-wave Trefftz spaces on polygonal meshes [17, 9]. Every plane-wave edge product is integrated analytically from

$$\int_e e^{ik\xi\cdot x}\,ds = |e|e^{ik\xi\cdot(x_a+x_b)/2}\,\operatorname{sinc}\left(\frac{k\xi\cdot(x_b-x_a)}{2}\right), \qquad \operatorname{sinc} t = \frac{\sin t}{t}, \qquad \operatorname{sinc}(0) = 1. \tag{52}$$

PWDG and UWVF are assembled independently at $\alpha = \beta = \delta = 1/2$. The UWVF assembly uses the impedance variables (8). By Theorem 1, its unscaled matrix satisfies $A_{\mathrm{U}}^{\mathrm{std}} = -2ikA_{\mathrm{P}}$, so scalar multiplication leaves the raw condition number unchanged. Likewise $G_{\mathrm{U}}^{\mathrm{std}} = 4kG_{\mathrm{P}}$, so the Riesz-normalized condition numbers coincide.

Table 1: Growth with local plane-wave dimension on the sixteen-triangle unstructured mesh. All edge integrals are analytic. The third column reports the raw PWDG condition number; the corresponding raw UWVF condition number is identical because $A_{\mathrm{U}}^{\mathrm{std}} = -2ikA_{\mathrm{P}}$. The fourth column is graph-normalized; the corresponding UWVF graph-normalized value is also identical because $G_{\mathrm{U}}^{\mathrm{std}} = 4kG_{\mathrm{P}}$.

p	$N = 16p$	$\kappa_2(A_{\mathrm{P}})$	$\kappa_2(\widehat{A}_{\mathrm{P}})$
5	80	1.57×10^{2}	2.917
9	144	9.07×10^{5}	6.172
13	208	5.31×10^{10}	9.789

The table also illustrates the distinction between raw and Riesz conditioning in a regime where the TDG Gram matrix is still numerically resolvable in double precision. Increasing the local plane-wave count from $p = 5$ to $p = 13$ increases the raw condition number by almost nine orders of magnitude, while the Riesz-normalized value grows only from about 2.92 to 9.79. At $p = 17$, the smallest eigenvalues of G_{P} lie at the double-precision noise floor and the untruncated inverse square root is no longer a reliable numerical operation; that row is therefore not reported. In practice one should either use

higher precision or state an explicit spectral truncation threshold and retained dimension, in line with the truncated local SVD/QR philosophy developed for UWVF [8].

The independent PWDG and UWVF assemblies also provide a direct check of Theorem 1. Over the three rows in Table 1,

$$\frac{\|A_{\mathrm{U}}^{\mathrm{std}} + 2ikA_{\mathrm{P}}\|_F}{\|A_{\mathrm{U}}^{\mathrm{std}}\|_F} \le 4.1 \times 10^{-16}, \qquad \frac{\|G_{\mathrm{U}}^{\mathrm{std}} - 4kG_{\mathrm{P}}\|_F}{\|G_{\mathrm{U}}^{\mathrm{std}}\|_F} \le 3.7 \times 10^{-16}. \tag{53}$$

These are the largest relative Frobenius residuals over $p = 5, 9, 13$; they are observed roundoff-level discrepancies, not prescribed tolerances. This independent assembly check is the numerical verification of the matrix equivalence.

To isolate the effect of the classical left scaling, we also assemble the UWVF directly as $A_{\mathrm{U}}^{w} = D - C$ in the weighted impedance basis and compare the matrices in (45). The same analytic edge integrals are used; no matrix in the table is inferred from the PWDG relation.

Table 2: Conditioning of the two classical UWVF reformulations and the graph–Riesz form on the same sixteen-triangle mesh. Here $A_L = I - D^{-1}C$, $A_T = D^{-1/2}(D-C)D^{-1/2}$, and $\widehat{A}_{\mathrm{U}} = G_{\mathrm{U}}^{-1/2}(D-C)G_{\mathrm{U}}^{-1/2}$, with $G_{\mathrm{U}} = (D-C) + (D-C)^*$.

p	$\kappa_2(D)$	$\kappa_2(A_L)$	$\kappa_2(A_T)$	$\kappa_2(\widehat{A}_{\mathrm{U}})$
5	7.68×10^{1}	2.55×10^{1}	10.366	2.917
9	4.12×10^{5}	5.49×10^{3}	10.910	6.172
13	2.29×10^{10}	4.65×10^{7}	18.702	9.789

The one-sided form substantially reduces the condition number of the raw UWVF matrix, consistent with its historical use as a local block preconditioner [5, 6], but it still inherits severe Euclidean distortion from the trace Gramian: between $p = 5$ and $p = 13$, $\kappa_2(D)$ grows by more than eight orders of magnitude and $\kappa_2(A_L)$ grows from about 26 to 4.7×10^7. In contrast, the symmetric trace-Riesz matrix remains below 20 throughout these rows. The graph–Riesz normalization is smaller still, because it removes not only the trace-coordinate metric but also uses the global transmission-defect metric generated by the UWVF/PWDG stability identity.

8.1. Dependence on wavenumber and mesh size

The graph-normalized condition number is basis invariant, but Proposition 2 shows that it need not be independent of k, h, or the global number of coupled elements. To separate these effects from high-order local basis degeneration, we use $p = 3$ and no rank truncation. Uniform subdivision of the sixteen-element mesh gives 64 and 256 triangles without changing element shapes.

Table 3: Wavenumber and mesh dependence with $p = 3$. The first block holds the mesh fixed. The second keeps $kh_{\max}$ fixed while refining the mesh and increasing k. The third refines at fixed $k = 4$.

family	N_K	k	$kh_{\max}$	$\kappa_2(D)$	$\kappa_2(D - C)$	$\kappa_2(\widehat{A}_{\mathrm{U}})$
fixed h	16	4	2.064	4.24	16.94	2.219
	16	8	4.128	1.81	10.48	2.046
	16	16	8.256	1.84	9.75	1.872
	16	32	16.512	1.54	9.36	1.843
fixed kh	16	4	2.064	4.24	16.94	2.219
	64	8	2.064	4.24	33.05	4.439
	256	16	2.064	4.24	69.30	7.177
fixed k	16	4	2.064	4.24	16.94	2.219
	64	4	1.032	6.76	40.57	4.446
	256	4	0.516	7.86	86.21	9.914

The fixed-mesh sweep shows that the graph condition number of this low-order discrete operator does not grow merely because k is increased on the same coarse mesh; in fact it decreases slightly. This statement concerns the algebra of the discrete operator; the increasingly under-resolved fixed mesh is not used to assess approximation accuracy. The refinement blocks show the complementary effect. At fixed local $kh_{\max}$, the graph-normalized condition number rises from 2.219 to 7.177 as the global coupled problem grows from 16 to 256 elements, and at fixed $k = 4$ it rises to 9.914. In this low-order refinement regime the raw and graph-normalized condition numbers grow on comparable scales: the raw-to-graph ratio stays between about 5 and 10. Coordinate contamination is therefore modest here. The dramatic separation established by this paper occurs under high-order enrichment and direction coalescence, where that ratio grows by many orders of magnitude. This is the behavior predicted by Proposition 2: after basis effects are removed, $\kappa_2(\widehat{A})$ records the continuity and inf–sup properties of the assembled operator.

8.2. Coalescing directions: isolating basis degeneracy

The preceding experiment increases the dimension of the local plane-wave space, so changes in approximation space and changes in basis conditioning occur simultaneously. We now hold the local dimension fixed at $p = 5$ and vary only one angular separation. On every element we take the directions

$$\theta = (0, \Delta\theta, 2.1, 3.5, 5.0),$$

with angles in radians. Thus two basis functions coalesce as $\Delta\theta \to 0$, while the remaining three directions and the global mesh are unchanged. The same analytic edge assembly is used throughout.

Table 4: Near-coalescing plane-wave directions with $p = 5$ on every element. Basis conditioning diverges as $\Delta\theta \to 0$, whereas the two symmetric operator condition numbers remain essentially constant.

$\Delta\theta$	$\kappa_2(D)$	$\kappa_2(A_L)$	$\kappa_2(A_T)$	$\kappa_2(\widehat{A}_{\mathrm{U}})$
10^{-1}	7.84×10^{3}	1.54×10^{3}	10.291	3.021
10^{-2}	8.08×10^{5}	1.61×10^{5}	10.290	3.029
10^{-3}	8.12×10^{7}	1.62×10^{7}	10.290	3.030
10^{-4}	8.12×10^{9}	1.62×10^{9}	10.289	3.030
10^{-5}	8.12×10^{11}	1.62×10^{11}	10.289	3.030

The scaling is decisive. Decreasing the angular separation by four decades increases $\kappa_2(D)$ by roughly eight decades, consistent with the rapid basis-conditioning deterioration previously documented for close or over-rich plane-wave sets [7, 8]. The classical one-sided matrix A_L inherits essentially the same divergence. By contrast, $\kappa_2(A_T)$ stays at about 10.29, and the graph-normalized condition number stays at about 3.03 to the displayed precision. The coupled operator condition numbers remain nearly unchanged while the plane-wave coordinate matrices deteriorate by many orders of magnitude. This gives a direct numerical separation of basis degeneracy from operator conditioning. The values in Table 4 are formed without spectral truncation through $\Delta\theta = 10^{-5}$; the 10^{-12} retained-range rule is introduced only in the finite-precision study below.

8.3. GMRES under coalescing coordinates

Because A_L and A_T are isospectral, eigenvalues alone cannot predict their Krylov behaviour; their similarity transform has condition number $\kappa_2(D)^{1/2}$. We therefore solve the same exact-wave systems by unrestarted GMRES to relative residual 10^{-10}, with zero initial guess and no truncation. Table 5 reports iteration counts. The counts are right-hand-side dependent, so they are used as a practical nonnormality check rather than as a definition of conditioning.

Table 5: GMRES iteration counts for the fixed exact-wave right-hand side in the coalescing-direction experiment. No spectral truncation is used.

$\Delta\theta$	raw $D - C$	one-sided A_L	trace A_T	graph $\widehat{A}_U$
10^{-1}	77	51	52	40
10^{-3}	64	45	52	40
10^{-4}	64	41	52	40
10^{-5}	64	36	52	40

The symmetric representations give essentially constant iteration counts as the two local directions coalesce: 52 iterations for A_T and 40 for the graph-normalized matrix in every displayed row. The raw and one-sided counts are not monotone despite their exploding Euclidean condition numbers. This illustrates the expected limitation of eigenvalue and condition-number arguments for nonnormal systems and shows why GMRES counts must be interpreted together with the right-hand side rather than used as a surrogate for the basis-independent operator condition number.

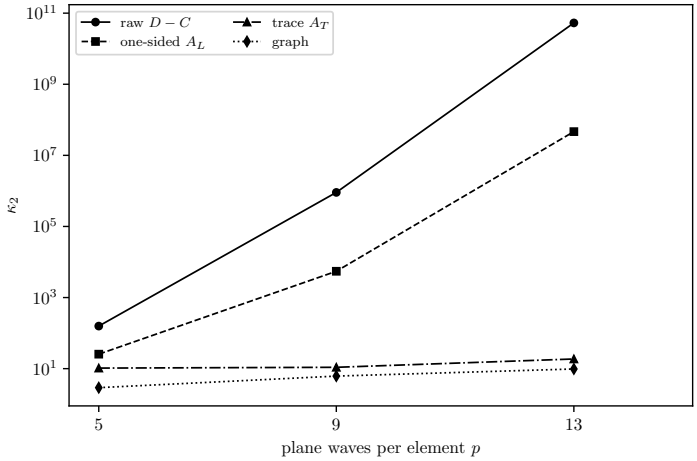


Figure 2: Condition numbers under local plane-wave enrichment. Raw and one-sided conditioning inherit the deterioration of the plane-wave coordinates, whereas the symmetric trace and graph representations remain moderate.

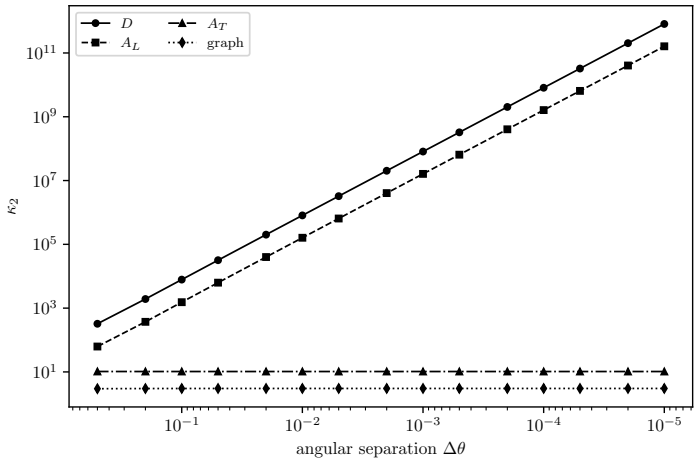


Figure 3: Condition numbers as two redundant local directions coalesce. The mesh, discrete dimension and Helmholtz problem are fixed; only the angular separation changes.

8.4. Finite-precision accuracy under basis collapse

Condition numbers alone do not show whether the distinction above affects a computed wave field. We therefore repeat the coalescing-direction experiment with an exact solution that is unchanged throughout the sweep. On every element the local space again has five plane waves, but the exact field is the global plane wave

$$u_{\mathrm{ex}}(x) = \exp\big(ikd(2.1) \cdot x\big),$$

whose direction 2.1 radians is one of the three fixed directions and is never moved. The two directions 0 and $\Delta\theta$ are redundant for representing this field. Hence the mesh, discrete dimension, differential equation, and exact discrete solution are all fixed; only the quality of the local coordinates deteriorates as $\Delta\theta \to 0$.

We solve the same discrete system in four coordinate representations: the raw UWVF matrix $D - C$, the classical left-scaled matrix A_L, the symmetric trace-scaled matrix A_T, and the graph-normalized matrix $\widehat{A}_{\mathrm{U}}$. For the two symmetric Riesz solves, eigenvalues below τ times the largest eigenvalue of the corresponding Gram matrix are discarded, with $\tau = 10^{-12}$ in Table 6. The tolerance specifies the numerical rank used in the discrete trace or graph metric. It is inactive while all Gram modes are resolved and removes a mode only when its energy in the relevant Gram metric falls below the chosen relative range. This has a field-level interpretation. The local impedance problem is stable: on each fixed Trefftz element there is a trace-to-volume bound of the form

$$\|v\|_{L^2(K)} \le C_K \|x_K(v)\|_{L^2(\partial K)}, \tag{54}$$

with the corresponding H^1 estimate under the standard impedance well-posedness assumptions; see the UWVF construction and Trefftz stability discussion in [1, 17]. A near-null mode of D_K therefore represents a coefficient combination with a small physical impedance trace and, after stable local reconstruction, a small element field. Truncation in the trace metric removes directions that are unresolved in the physical trace norm rather than directions chosen only by coefficient magnitude. The sensitivity study below shows a broad interval of tolerances with the same retained range and essentially the same field accuracy. Accuracy is measured in the reconstructed physical field by

$$E_{L^2} = \frac{\|u_h - u_{\mathrm{ex}}\|_{L^2(\Omega)}}{\|u_{\mathrm{ex}}\|_{L^2(\Omega)}}.$$

For comparison, we also compute the standard normwise relative backward residual [16]

$$\eta_{\mathrm{bw}} = \frac{\|Ax - b\|_2}{\|A\|_2\|x\|_2 + \|b\|_2}$$

for each representation.

Table 6: Finite-precision accuracy with a fixed exact plane wave and a collapsing redundant pair. r_D and r_G are the retained dimensions for the trace and graph Riesz solves at relative Gram tolerance 10^{-12}; $\eta_{\mathrm{bw}}^{\mathrm{raw}}$ is the normwise backward residual of the raw solve.

$\Delta\theta$	$\kappa_2(D)$	raw E_{L^2}	D^{-1} E_{L^2}	trace E_{L^2}	r_D	graph E_{L^2}	r_G	$\eta_{\mathrm{bw}}^{\mathrm{raw}}$
10^{-1}	7.84×10^{3}	3.58×10^{-15}	2.20×10^{-15}	9.68×10^{-16}	80	1.13×10^{-15}	80	7.68×10^{-17}
10^{-3}	8.12×10^{7}	1.89×10^{-13}	2.00×10^{-13}	7.52×10^{-14}	80	1.33×10^{-13}	80	7.16×10^{-17}
10^{-5}	8.12×10^{11}	1.81×10^{-11}	2.08×10^{-11}	8.43×10^{-12}	80	4.29×10^{-12}	70	8.91×10^{-17}
10^{-6}	8.12×10^{13}	2.06×10^{-10}	1.54×10^{-10}	3.38×10^{-14}	64	3.20×10^{-14}	64	7.20×10^{-17}
5×10^{-7}	3.27×10^{14}	5.22×10^{-10}	4.26×10^{-10}	8.44×10^{-15}	64	8.10×10^{-15}	64	8.88×10^{-17}
10^{-7}	2.89×10^{16}	3.43×10^{-9}	2.62×10^{-9}	5.38×10^{-16}	64	4.94×10^{-16}	64	7.58×10^{-17}
5×10^{-8}	2.40×10^{17}	7.64×10^{-9}	6.47×10^{-9}	5.28×10^{-16}	64	3.95×10^{-16}	64	6.93×10^{-17}

The raw and one-sided systems progressively lose digits in the reconstructed field as $\kappa_2(D)$ grows, while the raw solve remains normwise backward stable at the level of 10^{-16}–10^{-17}. Thus a backward-stable coefficient solve need not produce a comparably accurate

Trefftz field when the coefficient-to-field map is nearly singular [16]. The activation of the retained-rank columns also makes clear that the dramatic recovery at the smallest angular separations involves regularization in addition to a change of metric. At $\Delta\theta = 10^{-5}$ the trace Gramian still retains all 80 modes while the global graph Gramian retains 70. The missing ten are global transmission-defect modes whose graph energy has crossed the relative threshold; because G_U couples neighbouring elements, their count need not equal the sixteen local direction pairs.

To isolate those two effects, Table 7 compares the Riesz-range solves with Euclidean truncated SVD (TSVD) applied directly to the raw and one-sided matrices at the same relative tolerance, and with untruncated symmetric solves whenever the Gram matrices remain positive in double precision.

Table 7: Regularization controls at relative tolerance 10^{-12}. Euclidean TSVD of the raw matrix recovers most of the field accuracy once the redundant pair becomes unresolved. The one-sided matrix remains strongly coordinate distorted, and its Euclidean TSVD is less effective. Untruncated symmetric solves lose accuracy at the same stage.

$\Delta\theta$	raw	raw TSVD	D^{-1}	D^{-1} TSVD	trace trunc.	graph trunc.	trace no trunc.	graph no trunc.
10^{-4}	2.25×10^{-12}	9.28×10^{-12}	1.76×10^{-12}	3.89×10^{-12}	1.04×10^{-12}	8.92×10^{-13}	1.28×10^{-12}	1.62×10^{-12}
10^{-6}	2.06×10^{-10}	3.83×10^{-14}	1.54×10^{-10}	8.33×10^{-10}	3.38×10^{-14}	3.20×10^{-14}	2.05×10^{-10}	1.54×10^{-10}
10^{-7}	3.43×10^{-9}	1.74×10^{-15}	2.62×10^{-9}	1.33×10^{-8}	5.38×10^{-16}	4.94×10^{-16}	–	–

The control changes the interpretation of the experiment in an important way. Regularization is the main mechanism that restores the field once the basis has crossed the floating-point resolution threshold; Euclidean TSVD of the raw matrix can already recover nearly all lost digits. At $\Delta\theta = 10^{-4}$ raw TSVD is slightly worse than the untruncated raw solve because the chosen Euclidean tolerance discards a mode that remains useful for this right-hand side. At 10^{-6} the one-sided TSVD is substantially worse than the untruncated one-sided solve, illustrating that Euclidean truncation of the nonsymmetric left-scaled system need not remove modes in the physically relevant trace metric. The role of the trace and graph metrics is more specific: they identify near-null directions in physically meaningful norms, provide canonical basis-independent operator condition numbers, and give a principled retained range tied to trace or transmission energy. At $\Delta\theta = 10^{-7}$ the smallest computed eigenvalues of both Gram matrices have crossed zero at roundoff, so an untruncated square root is no longer defined reliably in double precision; the reported symmetric condition numbers and solves in that regime are therefore always range-restricted. This experiment is not an approximation-convergence test, since the exact discrete field is identical in every row.

8.5. Penalty-parameter robustness

The graph separation is not confined to the one-half parameters. Keeping the same mesh, $k = 4$, $p = 5$ and $\delta = 1/2$, we assemble the PWDG matrix for both equal penalties and points on the exact one-way curve $4\alpha\beta = 1$. The graph matrix is always formed from $(A_{\mathrm{P}} - A_{\mathrm{P}}^*)/(2i)$, so this test directly checks the general positive-penalty theory of Section 5.

Table 8: General positive PWDG penalties. The final column is the relative L^2 error for the off-grid exact plane wave $u_{\mathrm{ex}} = e^{ikd(0.37)\cdot x}$, which is not contained in the five-direction local space. No graph modes are truncated.

family	(α, β)	ρ	$\eta_{\alpha,\beta}/k$	$\kappa_2(\widehat{A}_{\mathrm{P}})$	E_{L^2}
equal	$(0.30, 0.30)$	0.60	1.00	4.392	2.293×10^{-2}
equal	$(0.50, 0.50)$	1.00	1.00	2.917	2.232×10^{-2}
equal	$(0.70, 0.70)$	1.40	1.00	2.294	2.306×10^{-2}
equal	$(1.00, 1.00)$	2.00	1.00	1.898	2.515×10^{-2}
one-way	$(0.25, 1.00)$	1.00	0.50	4.428	3.233×10^{-2}
one-way	$(0.50, 0.50)$	1.00	1.00	2.917	2.232×10^{-2}
one-way	$(1.00, 0.25)$	1.00	2.00	3.102	1.659×10^{-2}

Across these parameter changes the graph-normalized condition number stays between 1.90 and 4.43, confirming that the normal form persists away from the classical one-half point. The accuracy column prevents a misleading interpretation of that algebraic improvement. Along the equal-penalty line, decreasing $\kappa_2(\widehat{A}_{\mathrm{P}})$ from 4.392 to 1.898 does not monotonically improve the field error; the smallest graph condition number in the table actually has a larger error than the one-half choice. On the one-way curve the case $(\alpha, \beta) = (1, 0.25)$ has the smallest field error despite a graph condition number above 3. The graph condition number therefore measures the algebra of the discrete operator in its natural norm and must be read alongside approximation accuracy when flux parameters are compared.

A solution outside the discrete space.. The off-grid plane wave in Table 8 also provides a control in which discretization error is present. At the classical parameters $(\alpha, \beta, \delta) = (1/2, 1/2, 1/2)$, the raw, one-sided, trace-normalized and graph-normalized systems reconstruct the same discrete field to roundoff when no truncation is used: all four give $E_{L^2} = 2.232 \times 10^{-2}$, and their coefficient vectors differ from the raw solution by less than 4×10^{-15} in relative Euclidean norm. Symmetric normalization therefore does not alter approximation error; its role is to expose and control the algebraic representation. The benefit becomes visible when the coordinate basis approaches numerical singularity and regularization is required.

8.6. Sensitivity to the numerical-rank tolerance

To expose the role of the range tolerance, we fix $\Delta\theta = 10^{-6}$ and repeat the symmetric solves while varying τ. Table 9 reports the retained dimensions and field errors. There is a wide stable plateau: for $10^{-10} \geq \tau \geq 10^{-13}$ both metrics retain the same 64-dimensional numerical range and return essentially the same field. Once τ is reduced below the unresolved Gram eigenvalues, near-null coordinate directions re-enter and the trace-space field error rises by several orders of magnitude.

Table 9: Sensitivity to the relative Gram-eigenvalue tolerance at $\Delta\theta = 10^{-6}$. r_D and r_G are the retained dimensions in the trace and graph metrics.

τ	r_D	trace-Riesz E_{L^2}	r_G	graph-Riesz E_{L^2}
10^{-10}	64	3.38×10^{-14}	64	3.20×10^{-14}
10^{-12}	64	3.38×10^{-14}	64	3.20×10^{-14}
3×10^{-13}	64	3.38×10^{-14}	64	3.20×10^{-14}
10^{-13}	64	3.38×10^{-14}	64	3.20×10^{-14}
3×10^{-14}	68	9.94×10^{-12}	64	3.20×10^{-14}
10^{-14}	79	1.02×10^{-10}	70	8.76×10^{-11}

This supports using the tolerance as an explicit numerical-rank parameter tied to floating-point resolution and the desired reconstructed-field accuracy. The value $\tau = 10^{-12}$ used in Table 6 lies inside the stable plateau rather than at a tuned transition point.

9. Computational implications and recommended practice

By Proposition 5, the element-block diagonal of the UWVF graph Gramian is exactly $2D$. The trace normalization is therefore the canonical block-local realization of the same Riesz construction used globally by the graph normalization.

The two symmetric normalizations have different computational roles. The trace matrix $D = \mathrm{blockdiag}(D_K)$ is element-block local, so $D^{-1/2}$ can be constructed independently on each element by a small Hermitian eigendecomposition or SVD; this is directly compatible with the local orthogonalization and rank-reduction strategies already used in UWVF and PWDG implementations [5, 7, 8]. The graph Gramian G_{P} is global but retains the block sparsity induced by element-edge coupling: applying G_{P} requires only the same local and neighbouring-element block interactions that appear in the TDG form. Its Euclidean condition number is nevertheless basis dependent and can be very large when plane-wave coordinates become nearly dependent, so an iterative matrix-function approximation should be applied on a locally orthogonalized or numerically retained range rather than to an unresolved raw coordinate representation. The exact $G_{\mathrm{P}}^{-1/2}$ used here defines the intrinsic discrete condition number; large-scale implementations may approximate its action through Krylov/Lanczos or rational/polynomial matrix-function techniques without forming a dense square root [14, 15]. The cost is then governed by sparse applications of G_{P} together with the convergence of the chosen matrix-function approximation on the retained spectral interval. Designing the best scalable approximation for Trefftz graph maps is a solver problem separate from identifying the conditioning measure studied here.

The algebra is dimension-independent. In three dimensions, polygonal edges are replaced by element faces and plane-wave directions lie on S^2, but the impedance traces, the jump identity, the positive graph form and the symmetric normalization are unchanged in structure. The practical challenges in three dimensions are stronger directional redundancy on spherical direction sets and larger face-trace blocks; the conditioning mechanism has the same structure. The same distinction between local trace coordinates and the coupled operator therefore applies to tetrahedral or polyhedral UWVF/PWDG discretizations; three-dimensional UWVF computations already exhibit the relevance of local block conditioning [6, 11].

For reporting and implementation, the numerical results support four practices. First, raw Euclidean condition numbers should be identified explicitly as coordinate-dependent quantities. Second, a symmetric trace-normalized condition number should be reported when the purpose is to remove local trace-basis distortion, and the graph-normalized condition number should be reported when the purpose is to measure the coupled Trefftz operator. Third, any numerical-rank reduction should state the relative Gram tolerance and retained dimension. Fourth, algebraic backward residuals should be supplemented by a field- or trace-level accuracy measure whenever nearly dependent Trefftz coordinates are present. These quantities answer different numerical questions and should not be collapsed into a single raw matrix condition number.

10. Conclusions and implications

The numerical results demonstrate that basis conditioning and operator conditioning are genuinely different numerical phenomena and must be reported separately in plane-wave Trefftz discretizations. This conclusion sharpens earlier observations that local plane-wave conditioning can dominate the assembled algebra [7, 8]. Both enrichment and angular coalescence can drive raw PWDG/UWVF condition numbers through many orders of magnitude without producing comparable growth in the symmetrically normalized operators. In the coalescing-direction experiment the contrast is particularly clean: the trace Gramian reaches condition numbers of order 10^{12} while the trace-normalized and graph-normalized operators remain near 10.29 and 3.03, respectively. A raw matrix condition number therefore cannot, by itself, be interpreted as conditioning of the discrete Helmholtz propagation problem.

The direct formula (12) is the simplest way to state the classical equivalence: at $\alpha = \beta = \delta = 1/2$, PWDG is the UWVF written in broken Trefftz fields [3, 17]. The left-hand side of the UWVF uses the outgoing and incoming impedance variables, while the equivalent PWDG description uses averages and the actual Dirichlet and normal-flux jumps. There is no additional skeleton unknown once the impedance traces are pulled back by the local Trefftz solutions.

Proposition 1 explains why the two interior penalties have the values they do. The two oriented impedance transmission conditions contain the same information as the pair

$$[\![u]\!]_N, \qquad [\![\nabla u]\!]_N.$$

Averaging the two oriented impedance products cancels the mixed value–flux terms and leaves precisely the two positive PWDG jump products. At the physical impedance, the normalization needed to produce these two channels is exactly $\alpha = \beta = 1/2$.

For general positive α and β, the same algebra gives a useful interpretation of the PWDG flux family. The ratio α/β chooses the impedance $k\sqrt{\alpha/\beta}$. The product $\alpha\beta$ controls the scale and the affine weighting of the forward and reverse orientations. On $4\alpha\beta = 1$ the reverse coefficient vanishes and the representation reduces to a one-way UWVF; away from that curve the exact representation is the fixed two-orientation combination (32). For $4\alpha\beta < 1$, the reverse coefficient is negative, so the combination should not be called a mixture or convex average.

The classical UWVF scaling of Section 7 sharpens this distinction. The historically standard one-sided matrix $I - D^{-1}C$ [5, 9] is isospectral to the symmetrically trace-normalized matrix but can have very different singular values when the local trace

Gramian D is ill-conditioned. Thus the classical D^{-1} scaling captures the correct generalized spectrum without, by itself, removing Euclidean coordinate distortion.

The Riesz identities provide the corresponding operator interpretation. In broken-field coordinates,

$$A_{\mathrm{P}} = H + iG_{\mathrm{P}},$$

with G_{P} the TDG jump-and-boundary Gram matrix. Riesz coordinates turn the dissipative part into the identity and give $K + iI$. In UWVF coordinates the corresponding trace Riesz map gives $\frac{1}{2}I + iK_{\mathrm{U}}$, which is exactly the fixed scalar $-i/2$ times the PWDG normal form. The different raw matrices therefore represent the same edge-coupling problem in two coordinate systems rather than two independent operator geometries. This identifies the intrinsic conditioning problem of PWDG and UWVF: after the appropriate Riesz normalization, arbitrary deterioration of the plane-wave coordinates is removed and the remaining singular values measure the coupled discrete propagation operator itself. The construction is related to operator preconditioning [4] and local UWVF boundary orthonormalization [5], but here the relevant Riesz maps are derived directly from the UWVF trace form and the PWDG coercivity identity, and their distinct roles are made explicit.

Code and data availability

The scripts and tabulated data used to generate the numerical results and figures are provided as ancillary material with the arXiv version of this manuscript and will accompany the journal submission.

References

[1] O. Cessenat, B. Després, Application of an ultra weak variational formulation of elliptic PDEs to the two-dimensional Helmholtz problem, SIAM J. Numer. Anal. 35 (1998) 255–299.

[2] O. Cessenat, B. Després, Using plane waves as base functions for solving time harmonic equations with the ultra weak variational formulation, J. Comput. Acoust. 11 (2003) 227–238.

[3] C.J. Gittelson, R. Hiptmair, I. Perugia, Plane wave discontinuous Galerkin methods: analysis of the h-version, ESAIM Math. Model. Numer. Anal. 43 (2009) 297–331.

[4] R. Hiptmair, Operator preconditioning, Comput. Math. Appl. 52 (2006) 699–706.

[5] T. Huttunen, P. Monk, J.P. Kaipio, Computational aspects of the ultra-weak variational formulation, J. Comput. Phys. 182 (2002) 27–46.

[6] T. Huttunen, M. Malinen, P. Monk, Solving Maxwell's equations using the ultra weak variational formulation, J. Comput. Phys. 223 (2007) 731–758.

[7] S. Congreve, J. Gedicke, I. Perugia, Numerical investigation of the conditioning for plane wave discontinuous Galerkin methods, in: F.A. Radu, K. Kumar, I. Berre, J.M. Nordbotten, I.S. Pop (Eds.), Numerical Mathematics and Advanced Applications—ENUMATH 2017, Lecture Notes in Computational Science and Engineering 126, Springer, Cham, 2019, pp. 493–500.

[8] H. Barucq, A. Bendali, J. Diaz, S. Tordeux, Local strategies for improving the conditioning of the plane-wave Ultra-Weak Variational Formulation, J. Comput. Phys. 441 (2021) 110449.

[9] T. Luostari, T. Huttunen, P. Monk, The ultra weak variational formulation using Bessel basis functions, Commun. Comput. Phys. 11 (2012) 400–414.

[10] T. Luostari, T. Huttunen, P. Monk, Improvements for the ultra weak variational formulation, Int. J. Numer. Methods Eng. 94 (2013) 598–624.

[11] E. Darrigrand, P. Monk, Combining the ultra-weak variational formulation and the multilevel fast multipole method, Appl. Numer. Math. 62 (2012) 709–719.

[12] R. Hiptmair, A. Moiola, I. Perugia, Trefftz discontinuous Galerkin methods for acoustic scattering on locally refined meshes, Appl. Numer. Math. 79 (2014) 79–91.

[13] L. Yuan, A combined scheme of the local spectral element method and the generalized plane wave discontinuous Galerkin method for the anisotropic Helmholtz equation, Appl. Numer. Math. 150 (2020) 341–360.

[14] N.J. Higham, Functions of Matrices: Theory and Computation, SIAM, Philadelphia, 2008.

[15] Y. Saad, Numerical Methods for Large Eigenvalue Problems, 2nd ed., SIAM, Philadelphia, 2011.

[16] N.J. Higham, Accuracy and Stability of Numerical Algorithms, 2nd ed., SIAM, Philadelphia, 2002.

[17] R. Hiptmair, A. Moiola, I. Perugia, A survey of Trefftz methods for the Helmholtz equation, in: Building Bridges: Connections and Challenges in Modern Approaches to Numerical Partial Differential Equations, Springer, 2016, pp. 237–279.
[18] S. Kapita, The PWDG–UWVF Equivalence Revisited, arXiv:2608.23822v1 (2026), preliminary version of the present manuscript.
[19] S. Kapita, Direction-Adaptive Plane-Wave Discontinuous Galerkin Methods for the Helmholtz Equation, arXiv:2608.18380 (2026).
[20] J. Coyle, N. Nigam, The whys and hows of conditioning of DG plane wave Trefftz methods: a single element, arXiv:2509.14500 (2025).
[21] A. Buffa, P. Monk, Error estimates for the ultra weak variational formulation of the Helmholtz equation, ESAIM Math. Model. Numer. Anal. 42 (2008) 925–940.
[22] B. Adcock, D. Huybrechs, Frames and numerical approximation, SIAM Rev. 61 (2019) 443–473.
[23] T. Betcke, L.N. Trefethen, Reviving the method of particular solutions, SIAM Rev. 47 (2005) 469–491.
[24] E. Parolin, D. Huybrechs, A. Moiola, Stable approximation of Helmholtz solutions in the disk by evanescent plane waves, ESAIM Math. Model. Numer. Anal. 57 (2023) 3499–3536.
[25] E. Perrey-Debain, Plane wave decomposition in the unit disc: convergence estimates and computational aspects, J. Comput. Appl. Math. 193 (2006) 140–156.